

Github Link: https://github.com/erictracc/ITEC6180_Homework_3_PM4PY

What is the most common sequence of activities from start to end? (variant)

- The most common variant with a count of 7994:
 1. Reception
 2. Nurse Admission
 3. Order Blood Tests
 4. Last Lab Tests Results
 5. Doctor Admission
 6. Discharge (Doctor)
 7. Discharge

What is the most common activity?

- The most common activities are tied at 55079 occurrences.:
 - Reception and Discharge.
 - This is because the log can capture only parts of a process, meaning we saw in the logs that some people were only discharged, or only part of reception, and there was incompleteness to the log. This can be because the log has a beginning and an end in terms of how much data is captured, with processes that have existed before, and continue to exist after. This can bring some noise to our data.

Time Perspective

Average Durations Measured in Flow Time from Shortest Activities to Longest Activities

Activity	Average Flow Time	Average Delays in Flow Time
Nurse admission	797.75	3075.17
Doctor admission	1047.51	3984.34
Order blood tests	1456.76	1192.77
Reception	1965.83	1496.97
Order imaging test	2067.35	3938.95
Last lab test results	2971.52	4910.96
Order consultant	3126.36	5273.54
Additional vitals	3968.83	10150.99

Discharge (Doctor)	4701.74	8510.92
Imaging	4961.36	8196.99
Imaging decrypting	5714.28	12764.49
Consultant	5797.12	15756.64
External exams	6578.72	26068.35
Order external exams	10575.06	16784.71

- The large discrepancy between average delays and average flow time is likely due to the large presence of infrequent variants
- The average flow time across all variants is 37900.88 and the average delay for them is 39396.56
- To calculate the average delays we assumed that the average flow time was the standard time to complete these tasks

While the longest single case is 221 days 13:46:00 the longest variants are:

Longest In Terms of Flow Time	Total Mean Flow Time	Longest In Terms of # of activities	Total Mean Flow Time
('Reception', 'Last lab tests results', 'Order Consultant', 'Consultant', 'External exams', 'External exams', 'Order Consultant', 'Consultant', 'Order Consultant', 'Order Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'External exams', 'Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'Consultant', 'Order Consultant', 'Consultant',	4826340 seconds ~ 55 days 20:39:00	('Reception', 'Nurse admission', 'Order blood tests', 'Last lab tests results', 'Doctor admission', 'Order Consultant', 'Order Consultant', 'Order Imaging test', 'Consultant', 'Imaging', 'Order Imaging test', 'Imaging', 'Imaging decrypting', 'Order Imaging test', 'Order Consultant', 'Consultant', 'Order Consultant', 'Order Consultant', 'Imaging decrypting', 'Consultant', 'Order Imaging test',	89400 seconds ~ 1 day 00:50:00
	# of Activities		# of Activities
	30		35

'Consultant', 'Order Consultant', 'Consultant', 'Nurse admission', 'Additional vitals', 'Doctor admission', 'Additional vitals', 'Discharge (Doctor)', 'Discharge')		'Imaging', 'Imaging decrypting', 'Order Consultant', 'Order Consultant', 'Imaging', 'Order Consultant', 'Order Consultant', 'Consultant', 'Consultant', 'Consultant', 'Imaging decrypting', 'Discharge (Doctor)', 'Consultant', 'Discharge')	
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The Most Infrequent Variants

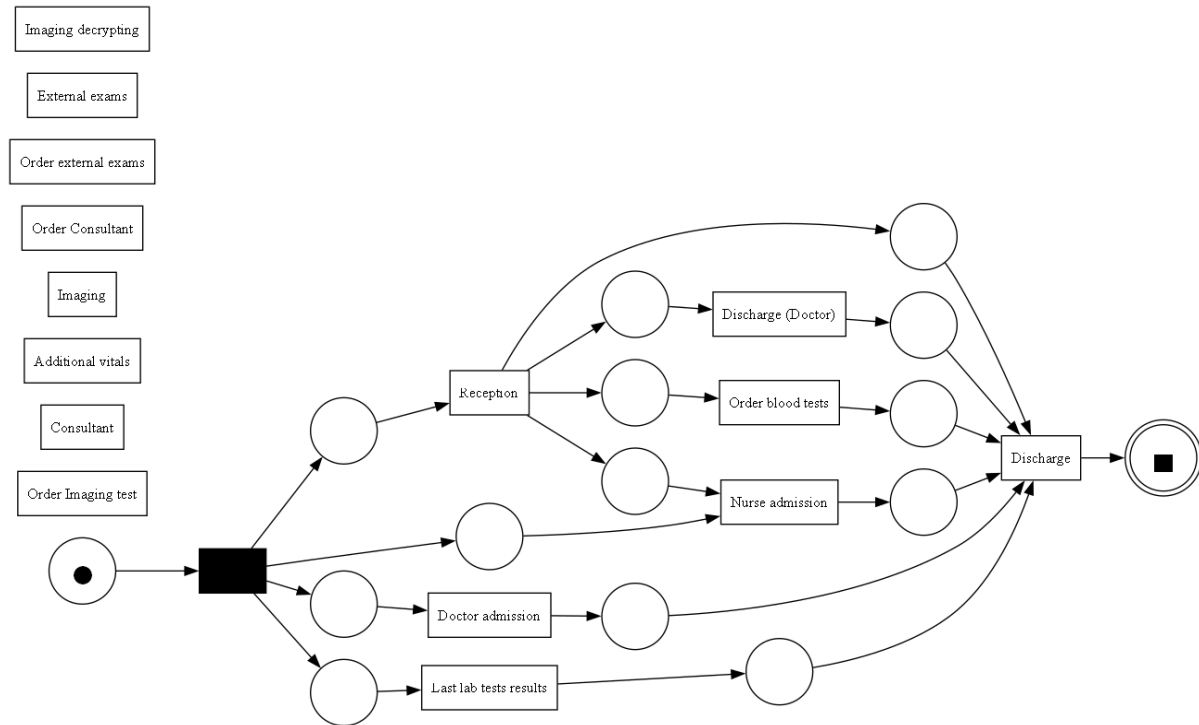
- There are 7703 variants which only occur once (these are probably all different variations of loops)

Are they the shorter or longer cases?

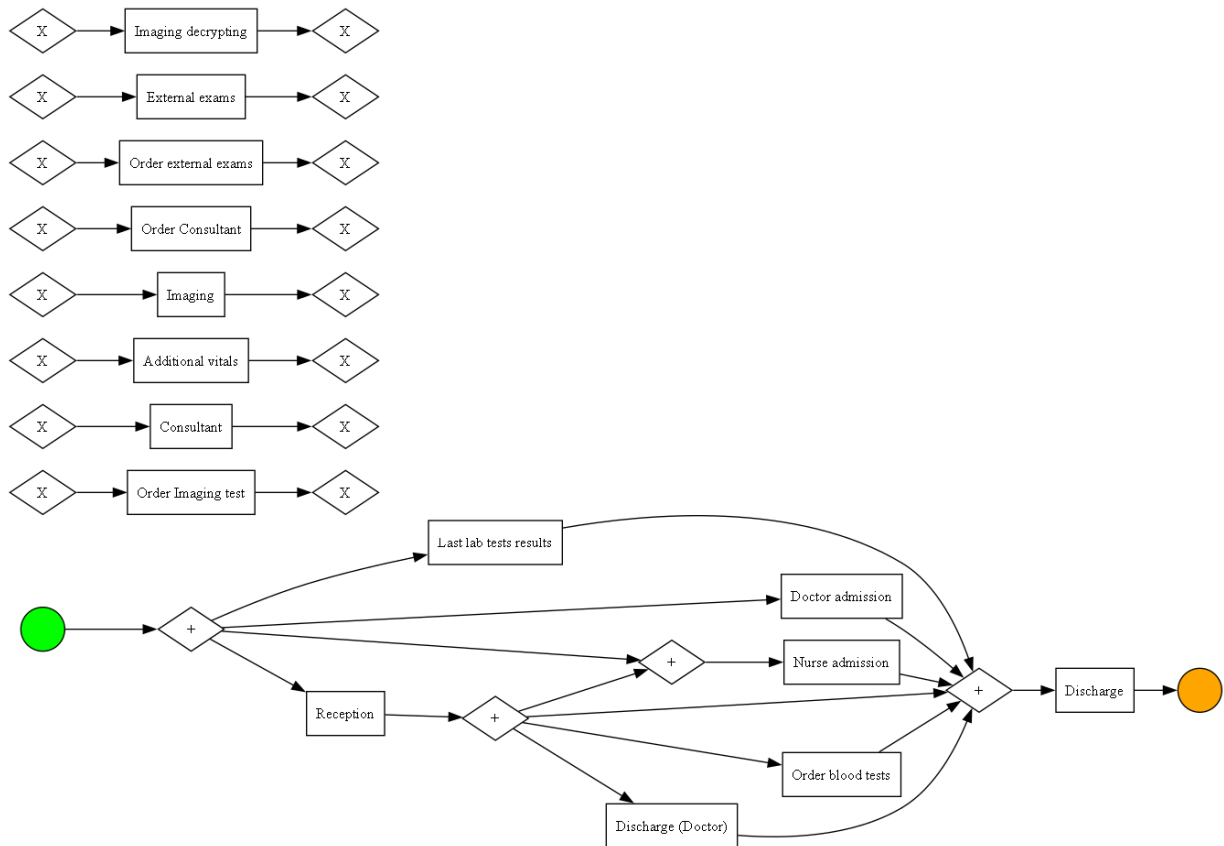
- The infrequent cases are the longer cases as they primarily consist of unusually long loops of activities
- Assumption: infrequent variants are those which occur once

Variants	Average Total Flow Time	Average # of Activities
Infrequent	40455.87	13.41
Frequent	27169.66	10.53

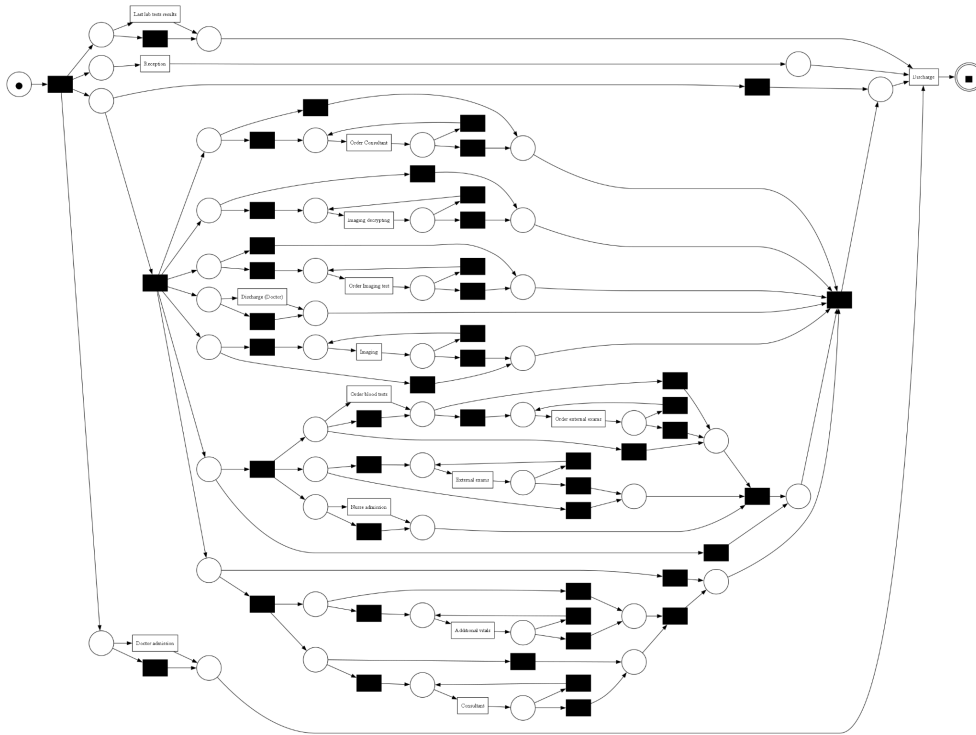
Alpha Miner + Models - Petri Net



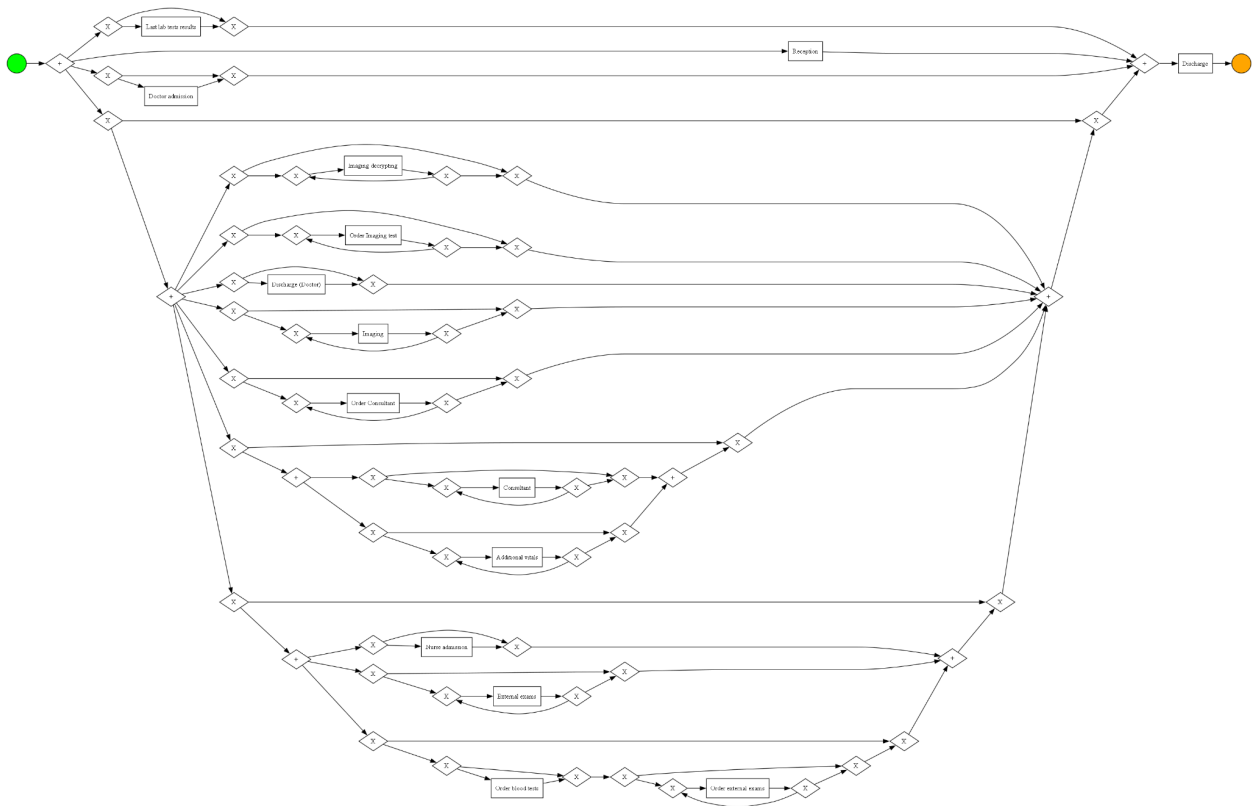
Alpha Miner + Models - BPMN



Inductive Miner Models - Petri Net



Inductive Miner Models - BPMN



Insights

- From Alpha + Miner
 - The generated model struggles to capture loops
 - The generated model includes captures more activities in the log than our BPMN model does
- From Inductive Miner
 - The generated model is able to capture loops, albeit in a more ambiguous manner
 - E.g. Order Consultant & Consultant activities are in visually separate flows when they would be better represented in the same flow
- From Heuristic Miner
 - The generated models capture every possible trace.
 - The generated DFG reveals a hanging discharge sequences/left censoring (unequal ingoing and outgoing)
- Other
 - Event log contains hanging sequences with sequence censoring, so we may not be getting the whole picture
- Usefulness
 - The models in general are able to better capture the activities in the event log than our BPMN
 - We can combine the missing activities with our interpretation of the model to better capture events that occur in the system.